ThinkWise: Learning When to Think in Large Multimodal Reasoning Models via Visual and Difficulty Perception

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https://github.com/Zhouwkk/ThinkWise

Abstract

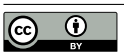
Large Multimodal Reasoning Models (LMRMs) extend large language models to heterogeneous modalities, demonstrating strong potential across a wide range of open-ended tasks. Reinforcement Learning with Verifiable Rewards (RLVR) has been widely adopted to improve reasoning performance in LMRMs; however, existing approaches overlook the critical role of visual perception uncertainty in guiding reasoning processes. We term this limitation Reasoning Allocation Misalignment, where the misalignment between visual perception uncertainty and reasoning depth leads to over-allocation of deep reasoning under uncertain visual grounding, resulting in unnecessarily long reasoning trajectories and degraded performance. To address this issue, we propose ThinkWise, a perception-aware RLVR framework that jointly models task difficulty and visual perception state to regulate reasoning depth. Specifically, we introduce an instance-adaptive visual perception modeling mechanism that enables context-aware calibration of perception uncertainty, leading to more accurate perception estimation. We further design a structured reward formulation that jointly considers perception, correctness, and task difficulty. In particular, an answer reward provides supervision for correctness, a perception- and difficulty-conditioned length reward adaptively regulates reasoning depth, and a perception reward encourages trajectories with stronger visual grounding. Extensive experiments on multiple multimodal reasoning benchmarks demonstrate that ThinkWise consistently improves the trade-off between accuracy and reasoning efficiency across both 3B and 7B models. These results highlight the importance of perception-aware reasoning and establish a principled framework for advancing LMRMs.

CCS Concepts

• **Computing methodologies** → **Knowledge representation and reasoning**.

Keywords

Large Multimodal Reasoning Models; Reinforcement Learning with Verifiable Rewards; Visual Perception



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MM '26, Rio de Janeiro, Brazil
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ACM ISBN 979-8-4007-2213-4/2026/11
https://doi.org/10.1145/3767308.3835837

1 Introduction

Large Multimodal Reasoning Models (LMRMs) extend the generative capabilities of large language models to heterogeneous modalities, enabling unified understanding and reasoning over text, images, and other signals, and thus holding great promise for a wide range of open-ended tasks [2, 10, 14, 19, 21, 26, 30, 39, 40, 43, 44, 46, 47, 50]. At the core of these capabilities lies multimodal reasoning, which can be naturally decomposed into two fundamental components: multimodal perception and logical reasoning [35, 49]. Prior studies [4, 11, 38] have shown that accurate visual perception is often the primary bottleneck, as it provides the essential visual basis upon which effective downstream reasoning critically depends.

Recent advances in Reinforcement Learning with Verifiable Rewards (RLVR) have demonstrated strong effectiveness in enhancing reasoning in large language models (LLMs), and have emerged as a dominant paradigm for improving multimodal reasoning capabilities of MLRMs by training them solely with rewards derived from the correctness of final answers [2, 21, 26, 26]. Given the critical role of visual perception, namely extracting task-relevant information from the image, more recent approaches have introduced dedicated mechanisms to strengthen this capability. One line of work [34, 50] enforces structured visual outputs to anchor perceptual grounding during reasoning. Another line [4, 11, 28] promotes visual re-attention to mitigate attention drift over long reasoning trajectories. The third line [30, 35] uses external visual perception rewards to alleviate reward sparsity and mitigate hallucinations.

Despite extensive efforts showing that improving visual perception enhances reasoning performance, optimizing perception in isolation remains insufficient for effective multimodal reasoning. A critical limitation lies in the failure to effective alignment visual perception quality and reasoning depth, which we term **Reasoning Allocation Misalignment**, where reasoning depth is allocated without considering whether a confident perceptual foundation has been established for the given task. In conventional LLM settings, models tend to allocate longer reasoning trajectories to more challenging problems, implicitly trading increased reasoning depth for improved accuracy. However, this situation becomes more complex in multimodal scenarios, where uncertainty in visual perception can introduce additional complexity. When reasoning depth is governed solely by task difficulty, models may over-allocate deep reasoning to cases with uncertain visual perception, leading to unnecessarily long trajectories and degraded performance. Although prior work [36] extends task difficulty to incorporate externally defined

measures of both problem complexity and visual perception difficulty, the absence of intrinsic perceptual assessment limits its effectiveness, and thereby highlighting the need for a more principled, perception-aware reasoning paradigm.

Building upon the above observations, a central challenge in multimodal reasoning is how to jointly model visual perception uncertainty and task difficulty to guide reasoning depth allocation. Addressing this requires moving beyond difficulty-driven strategies and explicitly incorporating perceptual uncertainty into the reasoning process. Our key insight is simple yet fundamental: deeper reasoning should be encouraged only when additional computation is both necessary and supported by sufficient visual involvement, namely when the task is difficult and the current trajectory remains strongly grounded in visual perception. Concretely, we design multiple **perception–correctness constraints** to regulate reasoning behavior. When visual perception is strong, incorrect trajectories, especially on difficult problems, are encouraged to continue reasoning to refine their responses, while correct trajectories are encouraged to remain concise to avoid unnecessary computation. In contrast, when visual perception is weak, deeper reasoning is discouraged regardless of correctness, as extended reasoning under insufficient visual support tends to amplify unreliable perceptual premises or reinforce shortcut-like solutions that are not grounded in the image. This perception-aware reasoning paradigm improves reasoning effectiveness by mitigating ineffective deep reasoning and reducing deductions caused by weakly visual perception, while also enhancing efficiency by avoiding redundant computation when deeper reasoning is unlikely to be beneficial.

To this end, we propose *ThinkWise*, a perception-aware RLVR framework for LMRMs. Our key innovation lies in moving beyond outcome-only supervision by explicitly incorporating trajectory-level perceptual signals into reasoning optimization, thereby enabling adaptive control over when deeper reasoning is beneficial. Specifically, we use Modality Attention Ratio (MAR) as a lightweight endogenous proxy, motivated by information-theoretic and internal-state perspectives [33], to estimate model's visual perception uncertainty. Building on this, we propose an instance-adaptive visual perception modeling mechanism that calibrates perception uncertainty in a context-aware manner, enabling more accurate perception signals. Finally, we propose a structured reward formulation that explicitly models perception–correctness constraints to guide reasoning behavior. The reward comprises three complementary components: an answer reward for correctness, a visual- and difficulty-perception length reward that adaptively regulates reasoning depth, and a perception reward that further promotes trajectories with stronger visual grounding. Through this design, ThinkWise elevates conventional outcome-only supervision into a structured, perception-aware training paradigm that enables principled reasoning allocation under perceptual reliability. We summarize our contributions as follows:

- We identify a fundamental limitation in multimodal reasoning, termed Reasoning Allocation Misalignment, where reasoning depth is allocated without considering visual perception, leading to suboptimal reasoning performance.
- We propose ThinkWise, a perception-aware RLVR framework that jointly models task difficulty and visual perception state to regulate reasoning depth, enabling adaptive reasoning allocation based on both necessity and perceptual reliability.
- We empirically evaluate ThinkWise on multiple MLLM reasoning benchmarks, demonstrating consistent improvements over strong RLVR baselines.

2 Preliminary

2.1 Problem Formulation.

We study multimodal reasoning in the context ofLarge Multimodal Reasoning Models (LMRMs). Let $\mathcal{D} = \{(v_i, q_i, a_i)\}_{i=1}^{N}$ denote a multimodal reasoning dataset, where v_i, q_i, and a_i represent the image, question, and reference answer of the i-th sample, respectively. Given an input pair $x_i = (v_i, q_i)$, the policy model (i.e., LMRM) π_θ generates a response sequence o_i that aims to reach the ground-truth answer a_i. This reasoning process can be decomposed into two stages: (1) multimodal perception, which extracts and grounds relevant information from inputs x_i, and (2) textual reasoning, which operates on the perceived information to perform step-by-step inference and produce the final answer. Crucially, the effectiveness of reasoning depends on the reliability of visual perception, making the alignment between perception and reasoning a key factor in multimodal reasoning.

2.2 Reinforcement Learning with Verifiable Rewards (RLVR).

RLVR is a training paradigm that optimizes a LMRM using rewards derived from verifiable outcomes. Given an input x_i, a policy model π_θ generates a response trajectory $o_i \sim \pi_\theta(\cdot \mid x_i)$, and a reward function $R(x_i, o_i)$ evaluates the correctness of the final output based on a task-specific verifier. The training objective is to maximize the expected reward:

$$\mathcal{L}_{\mathrm{RLVR}}(\theta) = \mathbb{E}_{x_i \sim \mathcal{D},\, o_i \sim \pi_\theta(\cdot|x_i)} \left[R(x_i, o_i) \right].$$

Group Relative Policy Optimization(GRPO). GRPO [25] is a widely used group-based policy optimization method in outcome-only RLVR. Formally, for each input x_i, we samples G candidate trajectories $\{o_i^j\}_{j=1}^{G}$ from the old policy model $\pi_{\theta_{\mathrm{old}}}$, and each trajectory is assigned a reward $R_i^j = R(x_i, o_i^j)$. GRPO first computes group-relative advantages by normalizing rewards within the sampled group:

$$\hat{A}_i = \frac{R_i^j - \mathrm{mean}(\mathbf{R})}{\mathrm{std}(\mathbf{R})}, \qquad \mathbf{R} = \{R_i^1, \ldots, R_i^G\}, \tag{1}$$

After obtaining the advantages, the policy model are then updated by maximizing the following objective:

$$J_(\theta) = \mathbb{E}_{x_i \in \mathcal{D},\, \{o_i^j\}_{j=1}^{G}} \left[\frac{1}{G} \sum_{j=1}^{G} \frac{1}{|o_i^j|} \sum_{t=1}^{|o_i^j|} \Big(g_{j,t}(\theta) - \beta\, D_{\mathrm{KL}}(\pi_\theta \,\|\, \pi_{\mathrm{ref}}) \Big) \right],$$
$$g_{j,t}(\theta) = \min \Big(r_{j,t}(\theta)\, \hat{A}_i,\ \mathrm{clip}(r_{j,t}(\theta), 1-\epsilon, 1+\epsilon)\, \hat{A}_i \Big). \tag{2}$$

where $r_{i,t}(\theta)$ denotes the probability ratio for the i-th trajectory at time step t, π_{ref} is the is a fixed reference model typically initialized from the initial policy.

In standard GRPO, the reward R_i for the i-th sampled output is typically defined solely by final-answer correctness, where $R_i \in$

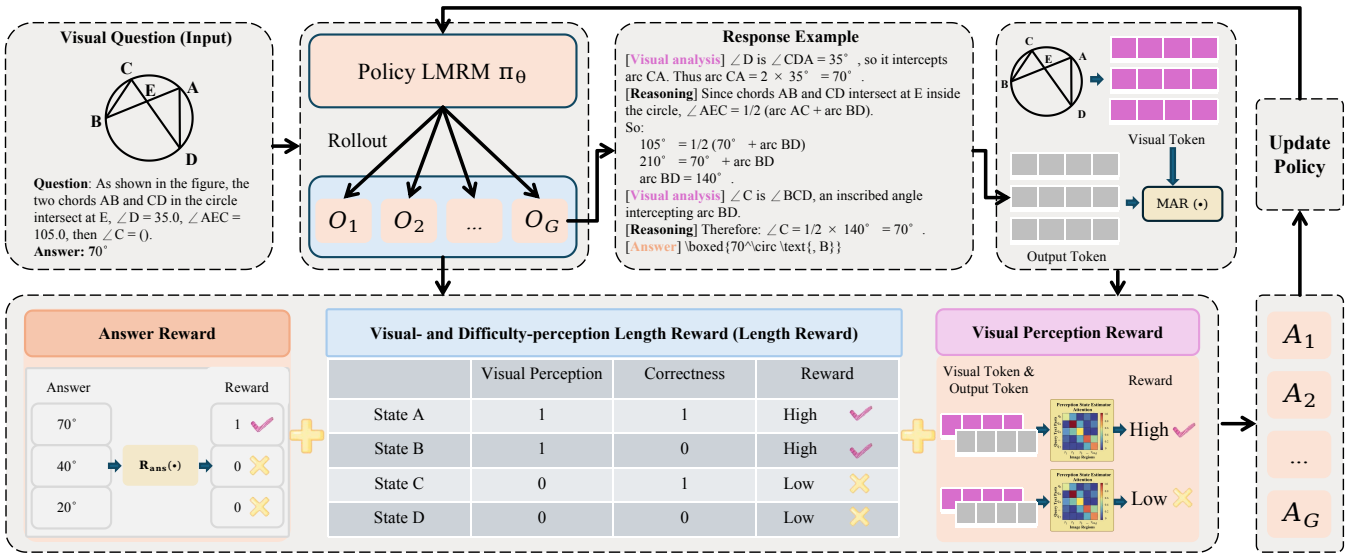


Figure 1: Overview of the ThinkWise training pipeline.

$\{0, 1\}$. While this sparse, outcome-driven supervision is highly effective for text-centric reasoning, relying strictly on it introduces a critical credit assignment problem in multimodal environments. In such settings, an incorrect response may originate from either flawed logical reasoning or inaccurate visual perception. Consequently, a binary outcome is insufficient to determine whether the model should be encouraged to allocate more computational resources (i.e., "think longer") to resolve the query. To address this limitation, ThinkWise preserves the core GRPO optimization objective while redesigning the sequence-level reward R_i to jointly account for visual perception uncertainty and task difficulty.

3 Methodology

In this section, we present the technical details of ThinkWise, a perception-aware RLVR framework that addresses reasoning allocation misalignment. The key innovation of ThinkWise lies in modeling trajectory-level perceptual uncertainty, jointly considering visual perception and task difficulty, to adaptively regulate reasoning depth beyond conventional outcome-only supervision. We first introduce the Modality Attention Ratio (MAR) as a lightweight and scalable endogenous proxy to quantify the grounding quality. To effectively use MAR for reward design, we discretize it into a binary perception state via an instance-adaptive threshold derived from group-wise statistics. This design avoids reliance on fixed thresholds, improves robustness across tasks with varying degrees of visual dependence, and provides a reliable, context-aware signal for perception-aware reasoning. Building on this visual perception signal, we design a structured reward mechanism that explicitly captures perception–correctness conditions to regulate reasoning behavior. The reward consists of three components: an answer reward for correctness, a *perception- and difficulty-conditioned length reward* that adaptively controls reasoning depth, and a visual perception reward that favors trajectories exhibiting strong visual perception. Together, these designs transform conventional outcome-only supervision into a structured, perception-aware training paradigm that aligns reasoning depth with both visual perception and task difficulty, enabling more effective multimodal reasoning.

3.1 Estimating Visual Perception uncertainty via the Modality Attention Ratio

To estimate the visual perception uncertainty of the current trajectory, we use the Modality Attention Ratio (MAR) as a proxy for visual engagement. Intuitively, stronger visual engagement corresponds to greater attention from output tokens to image tokens.

Formally, consider a model with L Transformer layers, and let $\mathcal{V}_{\text{img}}$ denote the set of image tokens. For the output token at position k (i.e., token in o_i) in layer l, we define its aggregated attention score over all visual tokens in Eq. (3):

$$s_k^{(l)} = \sum_{s \in \mathcal{V}_{\text{img}}} \mathbf{A}_{k,s}^{(l)}, \tag{3}$$

where $\mathbf{A}_{k,s}^{(l)}$ denotes the attention weight from position k to visual token s in layer l. To mitigate noise from shallow feature alignment and interference induced by deep language-dominant reasoning, we use a middle-layer subset $\mathcal{L}_{\text{mid}}$ by default and compute the average MAR within a computation window T, which is defined as:

$$\text{MAR}(o) = \frac{1}{|\mathcal{L}_{\text{mid}}|} \sum_{l \in \mathcal{L}_{\text{mid}}} \frac{1}{T} \sum_{t=1}^{T} s_t^{(l)}. \tag{4}$$

Note that $MAR(o)$ characterizes the model's average level of visual engagement during the early stage of the current trajectory.

3.2 Instance-Adaptive Visual Perception Modelling

To apply MAR as a control signal for reward allocation, we further discretize its continuous values into a binary visual perception

state. A straightforward approach is to adopt a global threshold. However, multimodal tasks exhibit substantial variability in their intrinsic dependence on visual information. For example, geometry reasoning typically requires strong visual involvement, whereas certain commonsense questions may rely less on visual information. Consequently, a single global threshold fails to generalize across instances (i.e., input pair x_i) and may misclassify valid trajectories in low-visual-dependency tasks as weakly visually engaged. In light of this, we leverage the group-wise sampling mechanism and introduce an instance-adaptive visual-engagement criterion. For G trajectories sampled from the same input x_i, we compute the group-wise mean and standard deviation of MAR, as defined: Eq. (5):

$$\mu_{\text{mar}}^{(i)} = \frac{1}{G}\sum_{j=1}^{G}\text{MAR}(o_i^j), \quad \sigma_{\text{mar}}^{(i)} = \sqrt{\frac{1}{G}\sum_{j=1}^{G}\left(\text{MAR}(o_i^j) - \mu_{\text{mar}}^{(i)}\right)^2}. \tag{5}$$

We define an instance-adaptive visual engagement threshold as a lower bound computed from group-level statistics, formulated as:

$$\tau_{\text{mar}}^{(i)} = \mu_{\text{mar}}^{(i)} - \alpha \cdot \sigma_{\text{mar}}^{(i)}, \tag{6}$$

where α is a hyperparameter controlling the strictness of the threshold. The final binary state assignment is given by Eq. (7):

$$e(o_i^j) = \mathbf{1}\left[\text{MAR}(o_i^j) > \tau_{\text{mar}}^{(i)}\right]. \tag{7}$$

Importantly, the threshold is defined relative to the MAR distribution within each sampled group, and the resulting visual perception state is thus interpreted as a context-aware control signal rather than an absolute measure of perceptual correctness. This design enables instance-adaptive discrimination of trajectories based on their relative visual engagement, identifying those that provide sufficient perceptual grounding to support deeper reasoning.

3.3 Perception-Conditioned Reward Design

With the carefully designed estimation methods for visual perception of each trajectory, we introduce ThinkWise, an RLVR framework that introduces a perception-conditioned reward formulation to explicitly incorporate visual perception and difficulty into the learning signal for guiding reasoning allocation, as illustrated in Algorithm 1. The overall reward consists of three components, each serving a distinct role in shaping reasoning behavior. Specifically, R_{ans} provides the base supervision for answer correctness; R_{len} adaptively modulates reasoning length conditioned on both visual perception uncertainty and task difficulty; and R_{perc} further encourages trajectories with stronger visual engagement beyond correctness alone. Together, these components enable a principled integration of perception and reasoning signals. The total reward is defined in Eq. (8):

$$R_{\text{total}}(o_i^j) = R_{\text{ans}}(o_i^j) + \gamma_{\text{len}} R_{\text{len}}(o_i^j) + \gamma_{\text{perc}} R_{\text{perc}}(o_i^j), \tag{8}$$

where γ_{len} and γ_{perc} are hyperparameters that balance the three terms.

3.3.1 Answer Reward. The base correctness reward is computed by a rule-based verifier that checks whether the final generated answer matches the reference answer under the predefined equivalence rule. Formally, the answer reward is defined in Eq. (9):

$$R_{\text{ans}}(o_i^j) = \begin{cases} 1, & \text{eq}(a_i, o_i^j) = 1, \\ 0, & \text{otherwise}, \end{cases} \tag{9}$$

where a_i denotes the reference answer, and $\text{eq}(a_i, o_i^j)$ is the answer equivalence function.

Table 1: Four constraints used in the visual- and difficulty-perception length reward. Here, $e(o) = 1$ and $e(o) = 0$ denote perception-strong and perception-weak trajectories.

State	Perception State	Correctness	Length-Reward Behavior
A	$e(o) = 1$	$R_{\text{ans}}(o) = 1$	Favor concise reasoning and suppress unnecessary verbosity.
B	$e(o) = 1$	$R_{\text{ans}}(o) = 0$	Encourage longer reasoning; the incentive is stronger on more difficult problems.
C	$e(o) = 0$	$R_{\text{ans}}(o) = 1$	Discourage overly long reasoning, as this may reflect shortcut behavior.
D	$e(o) = 0$	$R_{\text{ans}}(o) = 0$	Assign no additional length-based incentive.

3.3.2 Visual- and Difficulty-perception Length Reward (Length Reward). It is designed to regulate whether longer reasoning should be encouraged, rather than to uniformly favor either short or long responses. Its allocation depends jointly on three factors: the visual perception uncertainty of the current trajectory, answer correctness, and task difficulty. Intuitively, longer reasoning is encouraged only when the trajectory exhibits sufficient perceptual support yet still fails to produce the correct answer, especially on difficult problems. By contrast, when the answer is already correct or the visual perception is weak, additional length should be discouraged or left unrewarded. The four perception-aware constraints are summarized in Table 1. Formally, the length reward is defined in Eq. (10):

$$R_{\text{len}}(o_i^j) = \begin{cases} \text{clip}(1-\rho,\,-1,\,1), & e(o_i^j)=1,\ R_{\text{ans}}(o_i^j)=1 \\ d_i \cdot \text{clip}(\rho-1,\,0,\,1), & e(o_i^j)=1,\ R_{\text{ans}}(o_i^j)=0 \\ -\dfrac{1+d_i}{2} \cdot \text{clip}(\rho-1,\,0,\,1), & e(o_i^j)=0,\ R_{\text{ans}}(o_i^j)=1 \\ 0, & e(o_i^j)=0,\ R_{\text{ans}}(o_i^j)=0 \end{cases} \tag{10}$$

where $\rho = L(o_i^j)/L_{\text{avg}}$ denotes the relative reasoning length, with L_{avg} being the group-wise average trajectory length, and $d_i = 1 - \text{PassRate}_i$ is a difficulty factor derived from the pass rate of problem i. As shown in Table 1, d_i strengthens the incentive for extended reasoning in State B and scales the penalty on overly long reasoning in State C according to problem difficulty.

3.3.3 Visual Perception Reward. While the length reward regulates reasoning depth, it does not explicitly distinguish between correct trajectories supported by sufficient visual engagement and correct trajectories that may arise from shortcut-like behavior. To address this limitation, we introduce a separate perception reward R_{perc}, which is applied only to correct trajectories and further differentiates them according to their visual engagement.

Algorithm 1 ThinkWise Training

Require: Base model π_θ, selected dataset $\mathcal{D}$, group size G, truncation window T

Initialize $\pi_{\text{ref}} \leftarrow \pi_\theta$
for *epoch* = 1 to N **do**
  Sample batch $\{(v_i, q_i, a_i)\} \sim \mathcal{D}$
  Generate grouped trajectories $\{o_i^j\}_{j=1}^{G} \sim \pi_\theta(v_i, q_i)$
  Compute:
  • $\text{Acc}(o_i^j) = \text{eq}(a_i, o_i^j)$
  • $\text{MAR}(o_i^j)$ from multimodal attention over the first T reasoning tokens
  • $\tau_{\text{mar}}^{(i)} = \mu_{\text{mar}}^{(i)} - \alpha\sigma_{\text{mar}}^{(i)}$
  • $e(o_i^j) = \mathbf{1}[\text{MAR}(o_i^j) > \tau_{\text{mar}}^{(i)}]$
  Compute rewards:

$$R_{\text{total}}(o_i^j) = R_{\text{ans}}(o_i^j) + \gamma_{\text{len}} R_{\text{len}}(o_i^j) + \gamma_{\text{perc}} R_{\text{perc}}(o_i^j)$$

  • R_{len}: Eq. (10)
  • R_{perc}: Eq. (11)
  Update policy:

$$\max_\theta \mathbb{E}\Big[\text{GRPO}\Big(\pi_\theta, \pi_{\text{ref}}, \hat{A}, R_{\text{total}}\Big)\Big]$$

end for

Formally, the perception reward is defined in Eq. (11):

$$R_{\text{perc}}(o_i^j) = \begin{cases} +\gamma_1 \widetilde{\text{MAR}}(o_i^j), & e(o_i^j) = 1,\ R_{\text{ans}}(o_i^j) = 1, \\ -\gamma_2 \widetilde{\tau}_{\text{mar}}, & e(o_i^j) = 0,\ R_{\text{ans}}(o_i^j) = 1, \\ 0, & R_{\text{ans}}(o_i^j) = 0, \end{cases} \tag{11}$$

where $\widetilde{\text{MAR}}(o_i^j)$ and $\widetilde{\tau}_{\text{mar}}$ denote the group-normalized MAR score and threshold, respectively, as defined in Eq. (12).

$$\widetilde{\text{MAR}}(o_i^j) = \frac{\text{MAR}(o_i^j) - \text{MAR}_{\text{min}}}{\text{MAR}_{\text{max}} - \text{MAR}_{\text{min}}}, \quad \widetilde{\tau}_{\text{mar}} = \frac{\tau_{\text{mar}} - \text{MAR}_{\text{min}}}{\text{MAR}_{\text{max}} - \text{MAR}_{\text{min}}}, \tag{12}$$

where MAR_{min} and MAR_{max} are the minimum and maximum MAR values within the sampled trajectory group of the current input, and γ_1 and γ_2 control the strengths of the reward and penalty, respectively. For incorrect trajectories, $R_{\text{ans}}(o_i^j) = 0$ already provides the main supervision signal, so no additional perception reward is applied. For correct trajectories, R_{perc} distinguishes visually supported successes from perception-weak ones, discouraging shortcut-like solutions that bypass visual information.

4 Experiments

In this section, we conduct experiments to verify the effectiveness of ThinkWise, aiming to answer the following research questions:

- **RQ1:** How does ThinkWise compare with existing multimodal reasoning post-training methods on standard benchmarks?
- **RQ2:** What is the contribution of each key component in ThinkWise?
- **RQ3:** Does ThinkWise generalize across different RL algorithms?
- **RQ4:** How sensitive is ThinkWise to the main reward hyperparameters?

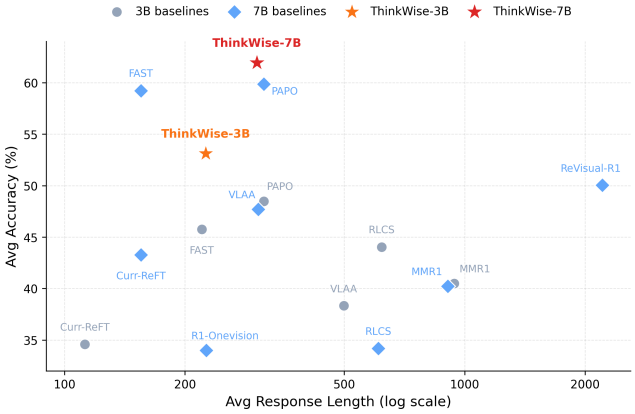


Figure 2: Accuracy versus average response length.

- **RQ5:** How does ThinkWise affect the model's reasoning behavior in representative cases?

4.1 Experimental Setup

4.1.1 Training Data and Protocol. All models are trained on ViRL39 [26] for one epoch with a learning rate of 1×10^{-6}. The dataset contains 38K verifiable multimodal reasoning samples covering visual mathematics, STEM reasoning, and chart understanding. To improve training stability, we further adopt a curriculum-style sampling strategy. In the early stage of training, we filter out overly easy samples so that the model focuses on sufficiently challenging examples and gradually improves its reasoning capability. In the later stage, we filter out overly difficult samples so that training concentrates on learnable cases, helping the model better learn when to reason deeply and when to respond concisely.

4.1.2 Baselines. We use Qwen2.5-VL-3B-Instruct and Qwen2.5-VL-7B-Instruct as the backbone models. We compare ThinkWise against three groups of baselines: the original base models, existing multimodal reasoning post-training methods, and generic RL algorithms (GRPO [25], Reinforce++ [9], and PPO [24]) together with their ThinkWise-based variants. The compared post-training methods include RLCS [32], VLAA [1], FAST [36], MMR1 [13], PAPO [31], Curr-ReFT [6], R1-OneVision [39], and ReVisual-R1 [3]. Since only 7B checkpoints are publicly available for R1-OneVision and ReVisual-R1, we include them only in the 7B setting.

4.1.3 Benchmarks and Metrics. We evaluate ThinkWise on benchmarks spanning mathematical and geometric reasoning, multidisciplinary multimodal reasoning, logical reasoning, and counting. MathVista [17], MathVerse [48], MathVision [29], WeMath [22], and MM-K12 [19] measure broad multimodal reasoning ability, while LogicVista [37], Counting[15], and MMMU-Pro [42] are used to assess visual grounding sensitivity. We report accuracy (Acc.) and average response length (Len.) for all benchmarks. All results are averaged over eight independent runs to account for output stochasticity.

Table 2: Main results on Qwen2.5-VL-3B and Qwen2.5-VL-7B. We report average accuracy (Acc.) and response length (Len.).

Method	MathVerse		MathVista		Counting		MMMU-Pro		MathVision		MMK12		We-Math		LogicVista	
	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓
Base Qwen2.5-VL-3B-Instruct																
Qwen2.5-VL-3B-Instruct	16.24	229.2	29.95	162.2	20.50	201.9	9.61	267.1	5.13	291.3	24.38	421.3	14.99	220.5	18.90	188.6
Curr-ReFT	38.11	98.3	52.47	88.5	43.91	82.6	18.91	106.1	8.58	125.3	40.52	209.1	31.06	105.0	34.65	103.2
RLCS	28.03	712.0	61.74	437.5	70.25	250.6	26.65	759.3	12.20	845.7	42.70	968.4	41.96	626.8	39.40	534.6
VLAA	39.79	718.1	51.19	332.2	55.25	185.8	21.99	381.6	9.13	877.8	40.21	673.2	43.02	660.3	32.47	388.6
FAST	39.93	287.3	**67.21**	146.3	72.50	82.5	22.49	274.8	**14.04**	309.0	47.38	339.2	38.53	216.8	36.05	157.3
MMR1	32.10	1078.9	56.40	644.3	47.06	315.2	27.17	1365.5	10.81	1347.8	39.01	1327.2	51.23	859.6	35.12	883.9
PAPO	**56.99**	353.9	58.35	184.4	57.69	101.5	26.57	401.2	11.50	420.9	57.12	529.6	59.35	348.3	**41.64**	242.3
ThinkWise-3B	56.82	262.6	59.16	153.2	**76.56**	109.0	**29.08**	215.2	12.32	327.1	**67.15**	342.6	**59.73**	258.5	41.11	167.7
Base Qwen2.5-VL-7B-Instruct																
Qwen2.5-VL-7B-Instruct	27.10	370.5	61.11	226.3	52.94	95.2	16.29	397.2	10.59	479.5	36.31	516.1	29.85	333.9	38.53	308.4
Curr-ReFT	37.60	147.8	63.48	129.0	62.25	97.7	19.35	182.4	10.93	202.3	47.57	241.3	44.70	134.6	42.25	160.5
RLCS	3.64	705.2	**70.00**	432.8	80.25	244.9	10.46	783.2	14.05	817.2	34.74	863.2	6.66	606.0	21.42	538.7
VLAA	43.93	329.0	63.49	216.9	75.75	95.3	24.43	367.8	12.41	441.4	44.67	492.1	47.79	309.3	43.26	286.8
FAST	66.78	146.2	64.41	115.2	78.56	72.7	34.80	231.0	13.74	284.7	76.33	174.2	66.08	130.0	**46.36**	162.7
MMR1	33.45	1058.0	59.36	627.4	72.50	257.2	16.68	1312.6	13.91	1391.9	24.32	1236.7	44.18	829.0	31.49	889.8
R1-Onevision	17.63	273.4	45.17	197.7	84.44	95.6	16.63	235.9	8.63	289.7	25.59	320.4	27.05	220.1	24.58	210.8
ReVisual-R1	55.35	2474.9	61.44	1529.1	67.19	890.2	31.66	2461.1	**16.93**	3369.9	37.36	3035.1	61.82	2135.1	40.10	2585.5
PAPO	**67.06**	378.6	66.79	213.0	87.25	94.6	34.81	398.4	13.80	465.9	72.32	441.4	66.19	327.3	44.74	284.7
ThinkWise-7B	66.07	344.7	67.45	240.8	**92.69**	122.7	**36.42**	348.1	14.84	410.7	**79.67**	428.3	**67.92**	306.1	44.94	275.9

4.2 Main Results (RQ1)

In Table 2, ThinkWise achieves the highest average accuracy across all eight benchmarks at both the 3B and 7B scales. Compared with the strongest baseline, PAPO, it improves the average accuracy by 4.09% and 2.13%, respectively, while producing responses that are 16.8% shorter on average across the two scales. Figure 2 further illustrates the accuracy–efficiency trade-off across all methods, showing that ThinkWise occupies the Pareto-optimal region at both scales.

4.2.1 Performance on perception-critical benchmarks. The largest improvements are observed on benchmarks that require reliable visual grounding for correct reasoning, especially Counting and MMK12, where ThinkWise achieves significant gains at both scales. These results are consistent with the hypothesis that explicitly coupling perception and reasoning leads to more effective use of visual information than outcome-only supervision.

4.2.2 Comparison with diffuculty-aware method. Compared with FAST, which considers both task and visual difficulty, ThinkWise performs better on tasks that require sustained visual engagement (e.g., MMK12 and We-Math), and also outperforms FAST on MathVerse at the 3B scale. FAST remains stronger on MathVista, where symbolic or spatial reasoning appears to more important than visual grounding. This pattern is consistent with the design of ThinkWise, whose perception-aware reward is most effective when visual grounding remains a bottleneck throughout the reasoning process.

4.2.3 Long-chain methods and ineffective deep reasoning. Long-chain methods such as MMR1 and ReVisual-R1 generate responses that are 3 to 8 times longer than those of ThinkWise, yet consistently obtain lower average accuracy. Without explicit alignment between perception and reasoning, longer reasoning chains may reinforce unreliable perceptual premises rather than improve the final answer.

4.2.4 Scaling behavior. The gains from 3B to 7B are most pronounced on Counting and MMK12, suggesting that tasks requiring broader visual understanding and stronger knowledge integration benefit more from increased model capacity. Across both scales, ThinkWise ranks first or second on nearly all benchmarks, indicating that the perception-aware design generalizes well across model sizes.

4.3 Ablation Study (RQ2)

Table 3 summarizes the ablation results of the three core components of ThinkWise on the 3B and 7B backbones.

4.3.1 Distinct roles of R_{perc} and R_{len}. Removing R_{perc} consistently reduces accuracy on both backbones, confirming that perception-aware reward is critical for LMRMs. Its effect on response length is scale-dependent: on 3B, removing R_{perc} leads to longer responses, suggesting that the smaller model is more prone to drifting into longer but weakly grounded reasoning traces without explicit groun ding-aware preference; on 7B, the length change is smaller and less uniform, indicating that the larger model already has stronger intrinsic grounding ability, so the main degradation appears in answer quality rather than a consistent shift in length.

Removing R_{len} increases response length on both scales, confirming that it is the primary mechanism for suppressing unnecessarily long outputs. The associated accuracy drop is generally smaller than that caused by removing R_{perc}, suggesting that R_{len} mainly regulates reasoning efficiency, whereas R_{perc} contributes more directly to reasoning quality. The stronger length increase on 3B further indicates that smaller models are more sensitive to length reward.

4.3.2 Effect of the curriculum. Removing the curriculum yields the most unstable behavior on both backbones, with especially severe length explosion on visually demanding tasks. In these cases, longer

Table 3: Ablation results on Qwen2.5-VL-3B and Qwen2.5-VL-7B.

Ablation	MathVerse		MathVista		Counting		MMMU-Pro		MathVision		MMK12		We-Math		LogicVista	
	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓
3B Ablation																
Qwen2.5-VL-3B-Instruct	16.24	229.2	29.95	162.2	20.50	201.9	9.61	267.1	5.13	291.3	24.38	421.3	14.99	220.5	18.90	188.6
ThinkWise-3B	**56.82**	262.6	59.16	153.2	**76.56**	109.0	**29.08**	215.2	**12.32**	327.1	**67.15**	342.6	**59.73**	258.5	**41.11**	167.7
w/o Length Reward	56.73	319.9	**61.34**	205.6	73.25	130.2	27.25	334.5	11.94	372.1	66.95	476	58.87	323.5	38.79	255.1
w/o Perception Reward	55.71	352.6	56.1	244.4	58.88	177.7	25.06	331.3	10.84	412.6	63.12	539.9	59.17	349	38.34	292.7
w/o Curriculum	55.42	650.6	55.31	592.8	56.69	187.1	23.74	645.9	11.45	2063.7	60.48	620.4	58.29	696.2	36.91	425.6
Vanilla GRPO	54.17	350.1	57.44	220.2	56.12	171.4	25.61	333.4	10.72	437.9	65.88	526.6	57.21	379.1	38.06	281.9
7B Ablation																
Qwen2.5-VL-7B-Instruct	27.10	370.5	61.11	226.3	52.94	95.2	16.29	397.2	10.59	479.5	36.31	516.1	29.85	333.9	38.53	308.4
ThinkWise-7B	**66.07**	344.7	**67.45**	240.8	**92.69**	122.7	**36.42**	348.1	**14.84**	410.7	79.67	428.3	**67.92**	306.1	44.94	275.9
w/o Length Reward	65.79	407.8	65.79	390.9	88.19	92.3	36.05	407.2	14.07	464.3	**80.74**	478.8	67.90	373.4	44.27	311.9
w/o Perception Reward	62.90	322.4	64.76	259.6	84.44	101.1	35.54	320.0	13.94	375.4	73.81	528.0	66.11	295.5	44.85	339.2
w/o Curriculum	63.72	500.3	64.05	454.5	80.81	199.8	35.60	672.3	12.56	2822.8	74.39	415.5	65.26	382.4	**45.86**	633.9
Vanilla GRPO	63.96	384.3	62.10	278.7	84.50	100.1	35.30	460.5	12.94	515.1	74.28	534.2	67.70	361.2	43.43	345.5

Table 4: Cross-algorithm generalization results on 3B models.

Optimizer	Variant	MathVerse		MathVista		Counting		MMMU-Pro		MathVision		MMK12		We-Math		LogicVista		Avg. Acc.	Avg. Len.
		Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓		
GRPO	Vanilla	54.17	350.1	57.44	220.2	56.12	171.4	25.61	333.4	10.72	437.9	65.88	526.6	57.21	379.1	38.06	281.9	45.65	337.58
	ThinkWise	56.82	262.6	59.16	153.2	76.56	109.0	29.08	215.2	12.32	327.1	67.15	342.6	59.73	258.5	41.11	167.7	50.24 (+4.59)	229.49 (-32.0%)
Reinforce++	Vanilla	46.79	317.2	43.10	211.5	41.50	117.0	19.13	306.3	4.70	93.9	50.55	377.6	52.01	291.7	35.57	246.6	36.67	245.23
	ThinkWise	57.72	304.0	60.16	188.9	76.00	142.2	26.74	305.5	13.82	349.3	70.41	361.1	60.14	284.5	41.30	236.8	50.79 (+14.12)	271.54 (+10.7%)
PPO	Vanilla	36.88	82.3	51.6	50.4	61	5.2	20.77	92.3	7.41	96.3	42.13	66.7	43.1	81.3	31.82	85.6	36.84	70.01
	ThinkWise	50.93	79.8	55.27	123.6	46.94	45	24.02	75.7	7.96	158.1	58.01	63	54.36	107.1	35.35	97.1	41.61 (+4.77)	93.68 (+33.8%)

reasoning does not improve performance, but instead amplifies errors from unreliable perceptual premises.

4.3.3 Comparison with vanilla GRPO.. Compared with vanilla GRPO, ThinkWise consistently improves accuracy on both model scales, with particularly clear gains on visually demanding benchmarks such as Counting, MMMU-Pro, and MMK12.

4.4 Generalization Across RL Algorithms (RQ3)

Table 4 evaluates the generalizability of ThinkWise across different RL algorithms on the 3B backbone. We consider three representative optimizers, namely GRPO, Reinforce++, and PPO. Across all settings, ThinkWise consistently outperforms the corresponding vanilla training, indicating that its effectiveness is not tied to any specific RL algorithm. Notably, the improvements are more pronounced on benchmarks that heavily rely on visual grounding (e,g., Counting, MMK12, MathVision, and MMMU-Pro), while relatively smaller gains are observed on tasks where visual perception plays a less critical role. Although accuracy improvements remain consistent across optimizers, the behavior of response length varies, suggesting that the accuracy benefits of ThinkWise are robust, whereas its impact on reasoning length is more sensitive to the choice of optimizer.

4.5 Sensitivity Analysis (RQ4)

Table 5 reports the results of three reward-weight settings on the 7B backbone, including the default setting S0 with $\gamma_{\text{len}} = 0.1$ and $\gamma_{\text{perc}} = 0.1$, a stronger length reward setting S1 with $\gamma_{\text{len}} = 0.2$, and a stronger perception reward setting S2 with $\gamma_{\text{perc}} = 0.2$. The results, shown in Table 5, indicate that S0 achieves the best trade-off between accuracy and response length among the three settings. The maximum average accuracy difference across all configurations is 2.30%, which suggests that ThinkWise remains reasonably stable under moderate changes in reward weights. We therefore use S0 as the default setting in all remaining experiments.

4.6 Case Study (RQ5)

To better understand how ThinkWise affects multimodal reasoning behavior, we annotate model failures. Specifically, for each model, we select 200 error cases from Counting, MMK12, MMMU-Pro, and We-Math, and categorize them into four types: ineffective deep reasoning, perception error, reasoning error, and calculation error. **Ineffective Deep Reasoning** occurs when the reasoning process drifts away from visual evidence and becomes dominated by language priors or incorrect premises. **Perception Error** arises when the model mmisreads key visual information and builds subsequent reasoning on a wrong perceptual premise. **Reasoning Error** refers to cases where visual perception is correct, but the reasoning process is flawed. Finally, **Calculation Error** occurs when both

Table 5: Sensitivity analysis of reward weights on 7B models.

Setting	MathVerse		MathVista		Counting		MMMU-Pro		MathVision		MMK12		We-Math		LogicVista		Avg. Acc.↑	Avg. Len.↓
	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓	Acc.↑	Len.↓		
S0 ($\gamma_{\text{len}} = 0.1$, $\gamma_{\text{perc}} = 0.1$)	**66.07**	344.7	67.45	240.8	92.69	122.7	**36.42**	348.1	**14.84**	410.7	**79.67**	428.3	**67.92**	306.1	44.94	275.9	**58.751**	309.66
S1 ($\gamma_{\text{len}} = 0.2$, $\gamma_{\text{perc}} = 0.1$)	62.31	308.2	**68.25**	199.1	91.00	97.5	33.50	255.9	12.27	348.8	75.65	329.4	65.32	265.0	43.32	222.1	56.453	**253.25**
S2 ($\gamma_{\text{len}} = 0.1$, $\gamma_{\text{perc}} = 0.2$)	65.34	353.6	68.09	246.1	**93.00**	115.8	36.06	365.8	14.18	410.6	79.60	429.9	66.58	304.9	**45.08**	290.0	58.491	314.59

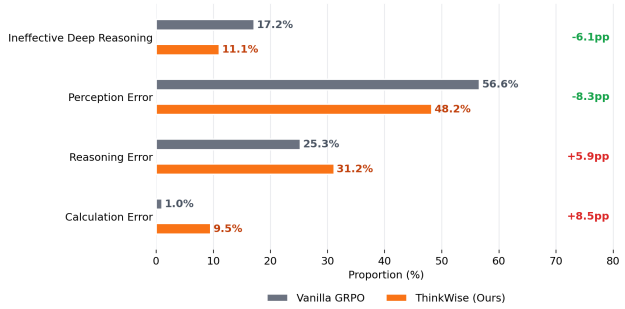


Figure 3: Annotated error category distributions.

perception and reasoning are correct, yet the final computation is incorrect. As shown in Figure 3, **ineffective deep reasoning** and **perception error** decrease by **6.1%** and **8.4%**, respectively, suggesting better use of visual evidence and less unproductive reasoning under weak grounding. By contrast, **reasoning error** and **calculation error** increase by **5.9%** and **8.5%**, respectively. This does not indicate weaker reasoning or numerical ability. Instead, once part of the perception-related failures is removed, the remaining errors become more concentrated in reasoning and computation. This suggests that ThinkWise shifts the primary bottleneck from visual grounding to how the model reasons and computes after identifying the relevant visual information.

5 Related Work

5.1 Reinforcement Learning with Verifiable Rewards.

RL-driven post-training has shown strong effectiveness for slow thinking in text-only large language models [8, 25, 41, 45], and is now being extended to multimodal reasoning. Recent works [10, 16, 19, 21, 39] advance multimodal reasoning post-training through data construction, RL, and cross-modal modeling. These studies show that RL-driven post-training improves multimodal reasoning while motivating adaptive reasoning control. A related line of work studies adaptive control of reasoning depth. DAST [27] reduces redundant reasoning on simple problems through difficulty-adaptive rewards, while O1-Pruner [18] mitigates length disharmony in long-chain reasoning through fine-tuning. Similar ideas have also been extended to multimodal settings. FAST [36] identifies overthinking in vision-language models and introduces adaptive thinking rewards to balance performance against token cost. These studies suggest that reasoning length can itself be treated as a controllable variable rather than a fixed byproduct of generation.

5.2 Large Multimodal Reasoning Models.

Effective reasoning in large multimodal reasoning models depends not only on logical deduction, but also on whether the reasoning process is grounded in correct perceptual evidence. Prior work [31] shows that current LMRMs are largely inherited from text-only settings and typically optimize reasoning using supervision based solely on final-answer correctness. However, such supervision can conflate two fundamentally different sources of failure, namely reasoning failure and perception failure. Wang et al. [31] further show that a substantial portion of multimodal reasoning errors arise from perception-related failures rather than logical reasoning errors. Meanwhile, Reich et al. [23] point out that answer correctness alone is insufficient to determine whether a prediction is genuinely grounded in visual evidence. Together, these findings suggest that final-answer supervision alone may be insufficient for modeling reasoning behavior in multimodal settings. Recent work has begun to explore internal model signals as a way to characterize grounding state. In LMRMs, grounding is closely related to internal attention behavior. iTaD [38] and subsequent studies [7, 20] report consistent correlations between attention-based features and grounding quality. At the same time, attention should not be interpreted as a direct measure of perceptual correctness, since attention weights alone do not fully explain Transformer behavior [5, 12].

6 Conclusion

In this work, we propose *ThinkWise*, a perception-aware RLVR framework that addresses reasoning allocation misalignment in large multimodal reasoning models. By incorporating trajectory-level perceptual signals via a lightweight endogenous proxy and introducing an instance-adaptive perception modeling mechanism, ThinkWise enables more accurate estimation of perceptual reliability. Building on this, we design a structured reward formulation that jointly models perception, correctness, and task difficulty to adaptively regulate reasoning depth. This transforms conventional outcome-only supervision into a perception-aware training paradigm, enabling more effective multimodal reasoning. Extensive experiments on multiple LMRM benchmarks demonstrate that ThinkWise consistently achieves significant improvements over strong RLVR baselines.

References

[1] Hardy Chen, Haoqin Tu, Fali Wang, Hui Liu, Xianfeng Tang, Xinya Du, Yuyin Zhou, and Cihang Xie. 2025. SFT or RL? An Early Investigation into Training R1-Like Reasoning Large Vision-Language Models. arXiv:2504.11468 https://arxiv.org/abs/2504.11468

[2] Liang Chen, Lei Li, Haozhe Zhao, Yifan Song, and Vinci. 2025. R1-V: Reinforcing Super Generalization Ability in Vision-Language Models with Less Than $3. https://github.com/Deep-Agent/R1-V. Accessed: 2025-02-02.

[3] Shuang Chen, Yue Guo, Zhaochen Su, Yafu Li, Yulun Wu, Jiacheng Chen, Jiayu Chen, Weijie Wang, Xiaoye Qu, and Yu Cheng. 2025. Advancing multimodal

reasoning: From optimized cold start to staged reinforcement learning. *arXiv preprint arXiv:2506.04207* (2025).

[4] Xu Chu, Xinrong Chen, Guanyu Wang, Zhijie Tan, Kui Huang, Wenyu Lv, Tong Mo, and Weiping Li. 2025. Qwen Look Again: Guiding Vision-Language Reasoning Models to Re-attention Visual Information. arXiv:2505.23558 https://arxiv.org/abs/2505.23558

[5] Kevin Clark, Urvashi Khandelwal, Omer Levy, and Christopher D. Manning. 2019. What Does BERT Look at? An Analysis of BERT's Attention. In *Proceedings of the 2019 ACL Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, Tal Linzen, Grzegorz Chrupała, Yonatan Belinkov, and Dieuwke Hupkes (Eds.). Association for Computational Linguistics, Florence, Italy, 276–286. doi:10.18653/v1/W19-4828

[6] Huilin Deng, Ding Zou, Rui Ma, Hongchen Luo, Yang Cao, and Yu Kang. 2025. Boosting the generalization and reasoning of vision language models with curriculum reinforcement learning. *arXiv preprint arXiv:2503.07065* (2025).

[7] Alessandro Favero, Luca Zancato, Matthew Trager, Siddharth Choudhary, Pramuditha Perera, Alessandro Achille, Ashwin Swaminathan, and Stefano Soatto. 2024. Multi-Modal Hallucination Control by Visual Information Grounding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 14303–14312.

[8] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, et al. 2025. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948* (2025).

[9] Jian Hu, Jason Klein Liu, Haotian Xu, and Wei Shen. 2025. REINFORCE++: Stabilizing Critic-Free Policy Optimization with Global Advantage Normalization. arXiv:2501.03262 https://arxiv.org/abs/2501.03262

[10] Wenxuan Huang, Bohan Jia, Zijie Zhai, Shaosheng Cao, Zheyu Ye, Fei Zhao, Zhe Xu, Xu Tang, Yao Hu, and Shaohui Lin. 2025. Vision-r1: Incentivizing reasoning capability in multimodal large language models. *arXiv preprint arXiv:2503.06749* (2025).

[11] Pu Jian, Junhong Wu, Wei Sun, Chen Wang, Shuo Ren, and Jiajun Zhang. 2025. Look Again, Think Slowly: Enhancing Visual Reflection in Vision-Language Models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng (Eds.). Association for Computational Linguistics, Suzhou, China, 9251–9270. doi:10.18653/v1/2025.emnlp-main.470

[12] Goro Kobayashi, Tatsuki Kuribayashi, Sho Yokoi, and Kentaro Inui. 2020. Attention is Not Only a Weight: Analyzing Transformers with Vector Norms. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu (Eds.). Association for Computational Linguistics, Online, 7057–7075. doi:10.18653/v1/2020.emnlp-main.574

[13] Sicong Leng, Jing Wang, Jiaxi Li, Hao Zhang, Zhiqiang Hu, Boqiang Zhang, Yuming Jiang, Hang Zhang, Xin Li, Lidong Bing, et al. 2025. Mmr1: Enhancing multimodal reasoning with variance-aware sampling and open resources. *arXiv preprint arXiv:2509.21268* (2025).

[14] Yunxin Li, Zhenyu Liu, Zitao Li, Xuanyu Zhang, Zhenran Xu, Xinyu Chen, Haoyuan Shi, Shenyuan Jiang, Xintong Wang, Jifang Wang, et al. 2025. Perception, reason, think, and plan: A survey on large multimodal reasoning models. *arXiv preprint arXiv:2505.04921* (2025).

[15] Zhuowan Li, Xingrui Wang, Elias Stengel-Eskin, Adam Kortylewski, Wufei Ma, Benjamin Van Durme, and Alan Yuille. 2023. Super-CLEVR: A Virtual Benchmark to Diagnose Domain Robustness in Visual Reasoning. arXiv:2212.00259 https://arxiv.org/abs/2212.00259

[16] Peng Liu, Peng-Fei Zhang, Jianfeng Qu, Ximing Li, Zhixu Li, and Pengpeng Zhao. 2026. Towards Practical LLM Unlearning: Efficient, Modular, and Retain-Free. In *Proceedings of the ACM Web Conference 2026*. 2846–2857.

[17] Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. 2023. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. *arXiv preprint arXiv:2310.02255* (2023).

[18] Haotian Luo, Li Shen, Haiying He, Yibo Wang, Shiwei Liu, Wei Li, Naiqiang Tan, Xiaochun Cao, and Dacheng Tao. 2025. O1-pruner: Length-harmonizing fine-tuning for o1-like reasoning pruning. *arXiv preprint arXiv:2501.12570* (2025).

[19] Fanqing Meng, Lingxiao Du, Zongkai Liu, Zhixiang Zhou, Quanfeng Lu, Daocheng Fu, Tiancheng Han, Botian Shi, Wenhai Wang, Junjun He, et al. 2025. Mm-eureka: Exploring the frontiers of multimodal reasoning with rule-based reinforcement learning. *arXiv preprint arXiv:2503.07365* (2025).

[20] Yuya Ogasa and Yuki Arase. 2025. Hallucinated Span Detection with Multi-View Attention Features. arXiv:2504.04335 https://arxiv.org/abs/2504.04335

[21] Yingzhe Peng, Gongrui Zhang, Miaosen Zhang, Zhiyuan You, Jie Liu, Qipeng Zhu, Kai Yang, Xingzhong Xu, Xin Geng, and Xu Yang. 2025. Lmm-r1: Empowering 3b lmms with strong reasoning abilities through two-stage rule-based rl. *arXiv preprint arXiv:2503.07536* (2025).

[22] Runqi Qiao, Qiuna Tan, Guanting Dong, Minhui Wu, Chong Sun, Xiaoshuai Song, Zhuoma GongQue, Shanglin Lei, Zhe Wei, Miaoxuan Zhang, Runfeng Qiao, Yifan Zhang, Xiao Zong, Yida Xu, Muxi Diao, Zhimin Bao, Chen Li, and Honggang Zhang. 2024. We-Math: Does Your Large Multimodal Model Achieve Human-like Mathematical Reasoning? arXiv:2407.01284 https://arxiv.org/abs/2407.01284

[23] Daniel Reich, Felix Putze, and Tanja Schultz. 2023. Measuring faithful and plausible visual grounding in VQA. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. 3129–3144.

[24] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347* (2017).

[25] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300* (2024).

[26] Haozhan Shen, Peng Liu, Jingcheng Li, Chunxin Fang, Yibo Ma, Jiajia Liao, Qiaoli Shen, Zilun Zhang, Kangjia Zhao, Qianqian Zhang, et al. 2025. Vlm-r1: A stable and generalizable r1-style large vision-language model. *arXiv preprint arXiv:2504.07615* (2025).

[27] Yi Shen, Jian Zhang, Jieyun Huang, Shuming Shi, Wenjing Zhang, Jiangze Yan, Ning Wang, Kai Wang, Zhaoxiang Liu, and Shiguo Lian. 2025. Dast: Difficulty-adaptive slow-thinking for large reasoning models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*. 2322–2331.

[28] Zhongwei Wan, Zhihao Dou, Che Liu, Yu Zhang, Dongfei Cui, Qinjian Zhao, Hui Shen, Jing Xiong, Yi Xin, Yifan Jiang, Chaofan Tao, Yangfan He, Mi Zhang, and Shen Yan. 2025. SRPO: Enhancing Multimodal LLM Reasoning via Reflection-Aware Reinforcement Learning. arXiv:2506.01713 https://arxiv.org/abs/2506.01713

[29] Ke Wang, Junting Pan, Weikang Shi, Zimu Lu, Mingjie Zhan, and Hongsheng Li. 2024. Measuring Multimodal Mathematical Reasoning with MATH-Vision Dataset. arXiv:2402.14804 https://arxiv.org/abs/2402.14804

[30] Peiyu Wang, Yichen Wei, Yi Peng, Xiaokun Wang, Weijie Qiu, Wei Shen, Tianyidan Xie, Jiangbo Pei, Jianhao Zhang, Yunzhuo Hao, et al. 2025. Skywork r1v2: Multimodal hybrid reinforcement learning for reasoning. *arXiv preprint arXiv:2504.16656* (2025).

[31] Zhenhailong Wang, Xuehang Guo, Sofia Stoica, Haiyang Xu, Hongru Wang, Hyeonjeong Ha, Xiusi Chen, Yangyi Chen, Ming Yan, Fei Huang, et al. 2025. Perception-aware policy optimization for multimodal reasoning. *arXiv preprint arXiv:2507.06448* (2025).

[32] Lai Wei, Yuting Li, Kaipeng Zheng, Chen Wang, Yue Wang, Linghe Kong, Lichao Sun, and Weiran Huang. 2025. Advancing Multimodal Reasoning via Reinforcement Learning with Cold Start. arXiv:2505.22334 https://arxiv.org/abs/2505.22334

[33] Hongxuan Wu, Yukun Zhang, and Xueqing Zhou. 2026. How Vision Becomes Language: A Layer-wise Information-Theoretic Analysis of Multimodal Reasoning. https://api.semanticscholar.org/CorpusID:285659939

[34] Jiaer Xia, Yuhang Zang, Peng Gao, Sharon Li, and Kaiyang Zhou. 2025. Visionary-R1: Mitigating Shortcuts in Visual Reasoning with Reinforcement Learning. arXiv:2505.14677 https://arxiv.org/abs/2505.14677

[35] Tong Xiao, Xin Xu, Zhenya Huang, Hongyu Gao, Quan Liu, Qi Liu, and Enhong Chen. 2025. Perception-R1: Advancing Multimodal Reasoning Capabilities of MLLMs via Visual Perception Reward. *arXiv preprint arXiv:2506.07218* (2025).

[36] Wenyi Xiao and Leilei Gan. 2025. Fast-slow thinking grpo for large vision-language model reasoning. *arXiv preprint arXiv:2504.18458* (2025).

[37] Yijia Xiao, Edward Sun, Tianyu Liu, and Wei Wang. 2024. Logicvista: Multimodal llm logical reasoning benchmark in visual contexts. *arXiv preprint arXiv:2407.04973* (2024).

[38] Xinhao Xu, Hui Chen, Mengyao Lyu, Sicheng Zhao, Yizhe Xiong, Zijia Lin, Jungong Han, and Guiguang Ding. 2025. Mitigating hallucinations in multimodal large language models via image token attention-guided decoding. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. 1571–1590.

[39] Yi Yang, Xiaoxuan He, Hongkun Pan, Xiyan Jiang, Yan Deng, Xingtao Yang, Haoyu Lu, Dacheng Yin, Fengyun Rao, Minfeng Zhu, et al. 2025. R1-onevision: Advancing generalized multimodal reasoning through cross-modal formalization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2376–2385.

[40] Huanjin Yao, Jiaxing Huang, Yawen Qiu, Michael K Chen, Wenzheng Liu, Wei Zhang, Wenjie Zeng, Xikun Zhang, Jingyi Zhang, Yuxin Song, et al. 2025. Mm-reason: An open-ended multi-modal multi-step reasoning benchmark for mllms toward agi. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 273–283.

[41] Qiying Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan, Gaohong Liu, Lingjun Liu, et al. 2025. Dapo: An open-source llm reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476* (2025).

[42] Xiang Yue, Tianyu Zheng, Yuansheng Ni, Yubo Wang, Kai Zhang, Shengbang Tong, Yuxuan Sun, Botao Yu, Ge Zhang, Huan Sun, et al. 2025. Mmmu-pro: A more robust multi-discipline multimodal understanding benchmark. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume*

1: Long Papers). 15134–15186.

[43] Peng-Fei Zhang, Guangdong Bai, and Zi Huang. 2026. Hierarchical Refinement of Universal Multimodal Attacks on Vision-Language Models. *IEEE Transactions on Multimedia* (2026).

[44] Peng-Fei Zhang, Guangdong Bai, Xin-Shun Xu, and Zi Huang. 2026. Data-Centric Challenges, Techniques, and Impacts: A Survey on Image Data Perturbation. *IEEE Transactions on Knowledge and Data Engineering* 38, 9 (2026), 5736–5764. doi:10.1109/TKDE.2026.3709786

[45] Peng-Fei Zhang, Ying Cheng, Xiaofan Sun, Shijie Wang, Fengling Li, Lei Zhu, and Heng Tao Shen. 2025. A step toward world models: A survey on robotic manipulation. *arXiv preprint arXiv:2511.02097* (2025).

[46] Peng-Fei Zhang, Zi Huang, and Guangdong Bai. 2024. Universal adversarial perturbations for vision-language pre-trained models. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 862–871.

[47] Peng-Fei Zhang, Yang Li, Zi Huang, and Xin-Shun Xu. 2022. Aggregation-Based Graph Convolutional Hashing for Unsupervised Cross-Modal Retrieval. *IEEE Transactions on Multimedia* 24 (2022), 466–479. doi:10.1109/TMM.2021.3053766

[48] Renrui Zhang, Dongzhi Jiang, Yichi Zhang, Haokun Lin, Ziyu Guo, Pengshuo Qiu, Aojun Zhou, Pan Lu, Kai-Wei Chang, Yu Qiao, et al. 2024. Mathverse: Does your multi-modal llm truly see the diagrams in visual math problems?. In *European Conference on Computer Vision*. Springer, 169–186.

[49] Jingqi Zhou, Sheng Wang, Jingwei Dong, Kai Liu, Lei Li, Jiahui Gao, Jiyue Jiang, Lingpeng Kong, and Chuan Wu. 2025. PROREASON: Multi-Modal Proactive Reasoning with Decoupled Eyesight and Wisdom. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. 31650–31679.

[50] Muzhi Zhu, Hao Zhong, Canyu Zhao, Zongze Du, Zheng Huang, Mingyu Liu, Hao Chen, Cheng Zou, Jingdong Chen, Ming Yang, et al. 2025. Active-o3: Empowering multimodal large language models with active perception via grpo. *arXiv preprint arXiv:2505.21457* (2025).